

Name of the experiment: Determination of sludge volume index (SVI) of given sample.

Principle:

SVI determination is based on estimating the volume of sludge settled in 30 minutes per gram of MLSS (mixed liquor suspended solids).

Apparatus:

- i) Imhoff cone or 1000ml measuring cylinder
- ii) Measuring cylinder,
- iii) crucibles,
- iv) beaker,
- v) Whatman filter paper no.40,
- vi) funnel, stand, hot air oven.

Procedure:

1. Take exactly one liter of a mixed liquor sample and allow it to settle in on Imhoff cone or measuring cylinder
2. Record the volume of settled sludge (V ml/l) at the end of 30 minutes.
3. Take the weight of a filter paper (W_o)
4. Take 100 ml sample after stir up and filter.
5. Dry the filter paper at 100-105°C until constant weight.
6. Take the final weight of the filter paper (W_f) and calculate MLSS.

Calculation:

$$MLSS = \frac{W_f - W_o}{\text{Sample volume in liter}}$$
$$= \text{mg/L}$$

From the obtained data, we can find the Sludge Volume Index (SVI) by following equation:

$$SVI \text{ (ml/g)} = \frac{V(SVI_{30}) \times 1000}{MLSS}$$

Result:

The amount of SVI determined from the provided sample is _____

Significance:

1. The value of SVI is of operational importance since it reflects changes in the treatment system.
2. SVI is used for determining the quality of sludge produced in a biological aeration unit and hence its efficiency.
3. It is used for determining the recirculation ratio necessary for maintaining a specified MLSS concentration in the aerator.
4. It is also used for estimating suspended solids concentration in recirculated sludge.